

Homework 1

Reading: Treil, Chapter 1, §§1–2.

Note. Do not use any conclusion outside Chapter 1, §§1–2, such as row reduction, determinants, rank, or dimension theorems.

[LADW] refers to the textbook *Linear Algebra Done Wrong* by Sergei Treil.

1. [LADW] Exercises 1.2(a–d).
2. [LADW] Exercises 1.7.
3. [LADW] Exercises 1.8.
4. [LADW] Exercises 2.1.
5. [LADW] Exercises 2.2.(a–e)
6. [LADW] Exercises 2.3.
7. [LADW] Exercises 2.5.
8. For $s \in \mathbb{R}$, define

$$\mathbf{v}_1 = (1, 0, 1)^T, \quad \mathbf{v}_2 = (0, 1, 1)^T, \quad \mathbf{v}_3 = (1, 1, s)^T.$$

- (a) Determine all s for which the ordered system $\mathcal{B}_s = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is a basis of $\mathbb{R}^3$. For every such s , find formulas for the coefficients in

$$(x, y, z)^T = \alpha \mathbf{v}_1 + \beta \mathbf{v}_2 + \gamma \mathbf{v}_3.$$

Explain why the formulas prove both existence and uniqueness.

- (b) Set $s = 3$. Find the coordinate vector of $\mathbf{w} = (2, -1, 4)^T$ with respect to $\mathcal{B}_3$. Verify your answer by substituting into the linear combination.
- (c) For every excluded value of s , describe exactly which $(x, y, z)^T$ are generated by the system, and give one vector that is not generated.

9. Let P_2 be the real vector space of polynomials of degree at most 2, including the zero polynomial. Define

$$q_-(t) = \frac{t(t-1)}{2}, \quad q_0(t) = 1 - t^2, \quad q_+(t) = \frac{t(t+1)}{2}.$$

- (a) Evaluate all three polynomials at -1 , 0 , and 1 . Use those values to prove that q_- , q_0 , q_+ are linearly independent.
- (b) Prove that every $p \in P_2$ satisfies

$$p(t) = p(-1)q_-(t) + p(0)q_0(t) + p(1)q_+(t).$$

Deduce that $\mathcal{Q} = (q_-, q_0, q_+)$ is a basis of P_2 .

- (c) For $p(t) = 2 - t + 3t^2$, find its coordinate vectors with respect to $\mathcal{Q}$ and to the standard ordered basis $(1, t, t^2)$. Explain why the coordinate vectors need not agree.